PORTFOLIO · 2026

Parts Cycle

How long parts take from the warehouse to the workshop, measured fairly enough to set a target on.



Givi Jananashvili · Product owner & developer

Internal web app for a premium automotive service centre · 11 staff users

THE PROBLEM

Everyone felt the delay. Nobody could measure it.

Requests went by email

No timestamps, no trail. A request could sit on a desk for an hour and leave no mark.

Delays were anecdotes

The warehouse and the service each blamed the other, and both were partly right.

No basis for a target

Management wanted a KPI. A KPI needs a number both sides accept as fair.

HOW IT WORKS

Four presses, three roles

01

Advisor

Sends the order

Request number and part count. The clock starts.

02

Warehouse

Starts collecting

The time before this press still counts as the warehouse's.

03

Warehouse

Dispatches the parts

The warehouse figure is final: request in, parts out.

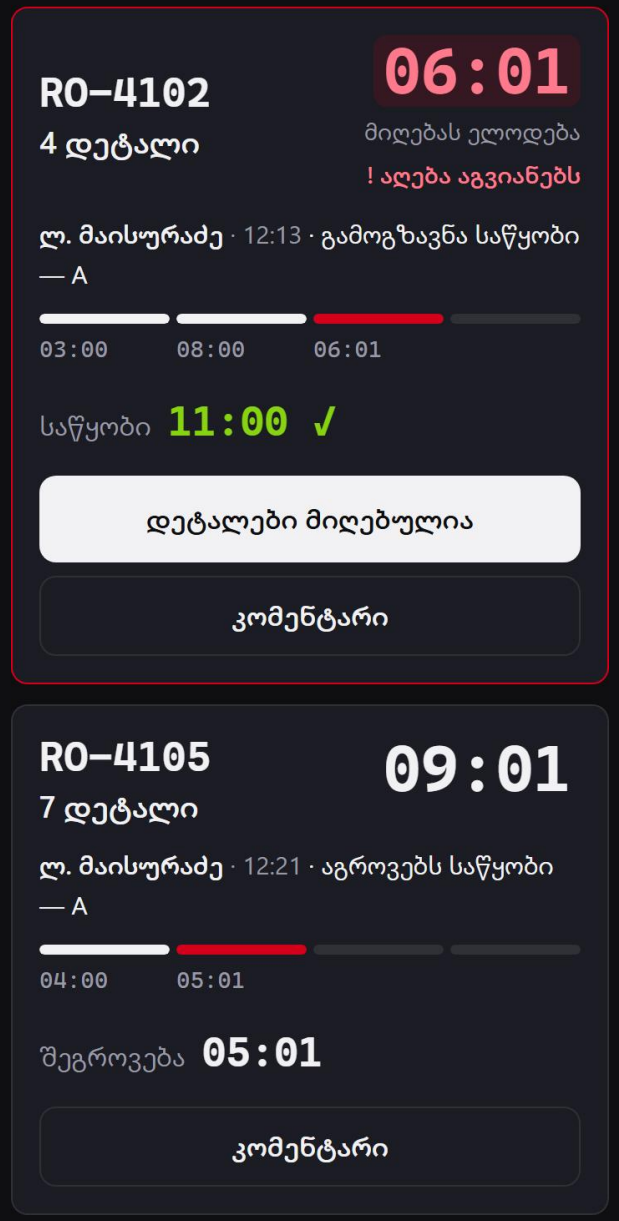
04

Receiver

Confirms receipt

The total stops. Pickup time is the receiver's own figure.

The parts list still goes by email. The app only measures time, so adopting it costs one tap per step.



HOW IT WORKS

The live board

Clocks count working time only

Sent at 18:40, picked at 09:10: ten minutes, not fourteen hours. Nights, Sundays and public holidays stop the clock.

Ownership changes at dispatch

The warehouse figure freezes in green. The big clock becomes the pickup wait: amber at 3 minutes, red at 5, and the receiver gets one notification.

Phone-first

Most of the 11 users work from phones, some in gloves: 48 px touch targets, every change checked at 390 px.

FAIR NUMBERS

Fairness by design

The measured side never sets its own yardstick

The advisor declares the part count, not the warehouse. It can be corrected only until the warehouse starts.

A skipped press is a missed target

Not pressing on a slow order can never make the numbers look better.

Tomorrow's orders carry no clock

Parts picked the day before for tomorrow's car stay out of every figure; nobody was waiting on them.

Holidays computed, not listed

Fixed public holidays plus Orthodox Easter, derived for any year. There is no list to forget to update.

FAIR NUMBERS

Two channels, one set of rules

Service

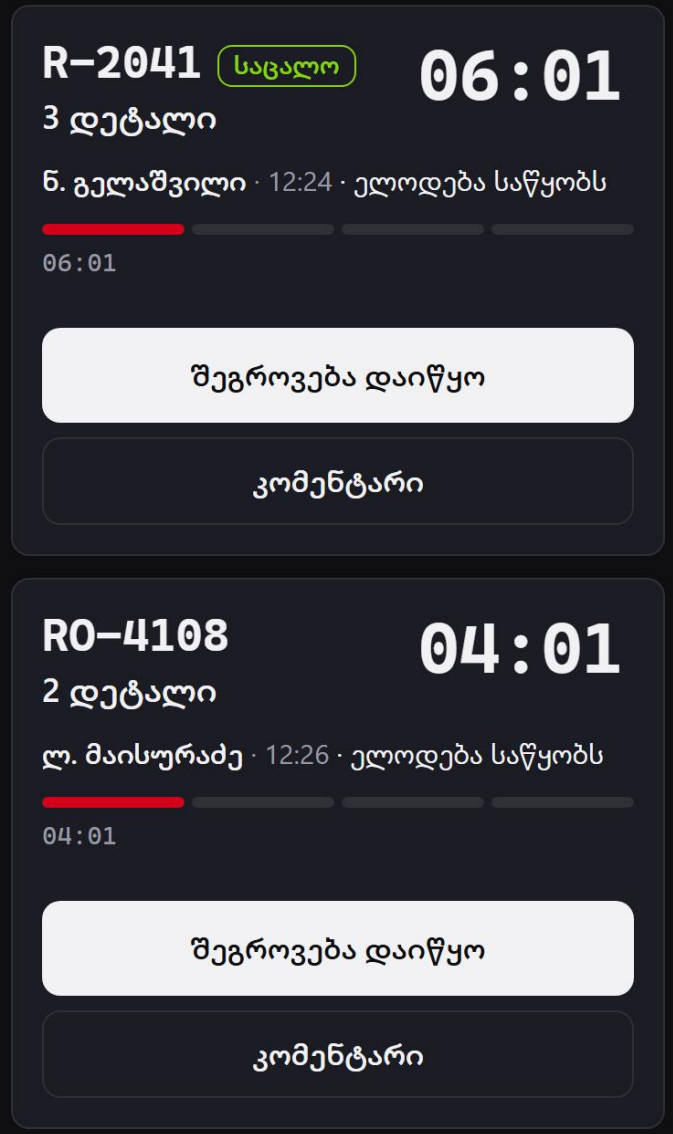
Advisors order for cars in the workshop; the service manager receives.

Retail counter

Parts advisors order for walk-in customers and receive the parts themselves.

Stamped once, reported apart

Each request gets its channel when sent and keeps it. The warehouse sees both, counter orders tagged in green; every figure is reported per channel.



THE KPI

Set the target on nine in ten, not on the average.

15 min

average, 1–5 parts

13 min

9 in 10 were out within this

A few stuck orders pulled the average above the time most requests met. A target on the average is unfair or disputed; on nine in ten it cannot be met by being quick on easy orders only.

Sample data

1–5 დეტალი	55 განაცხადი	
	საშუალოდ	10-დან 9
	15 წთ	13 წთ
ღილაკი გამოტოვა	0	

6–10 დეტალი	31 განაცხადი	
	საშუალოდ	10-დან 9
	26 წთ	21 წთ
ღილაკი გამოტოვა	0	

11–15 დეტალი	24 განაცხადი	
	საშუალოდ	10-დან 9
	30 წთ	27 წთ
ღილაკი გამოტოვა	0	

16–20 დეტალი	18 განაცხადი	
	საშუალოდ	10-დან 9
	31 წთ	33 წთ
ღილაკი გამოტოვა	0	

21+ დეტალი	12 განაცხადი	
	საშუალოდ	10-დან 9
	41 წთ	43 წთ
ღილაკი გამოტოვა	0	

THE KPI

What management reads

დრო ≈ **5.1 წთ** + **1.5 წთ** × დეტალების რაოდენობა

5 დეტალი	10 დეტალი
≈ 13 წთ	≈ 20 წთ
15 დეტალი	20 დეტალი
≈ 28 წთ	≈ 35 წთ

A formula that resists outliers

Fitted to the median time at each part count, weighted by volume. One stuck order cannot tilt the line.

Base minutes high

Requests sit unnoticed: an organisational fix.

Per-part minutes high

Parts are hard to find: a shelving fix.

Sample data · five part-count bands, weekly and monthly, per channel · Excel export for both channels

HOW IT IS BUILT

The architecture

Phones

Installable web app, vanilla JS. Service worker opens it instantly; Web Push for alerts.

Google Cloud Run

Python, FastAPI. Every timestamp is written by the database, never by a phone.

PostgreSQL

Supabase, own schema, row security on with no public policies. Append-only event log.

210

automated tests, end-to-end against a real PostgreSQL

7

schema migrations, always applied before the code ships

~3 min

from a push to main to live, via Cloud Build

HOW IT IS BUILT

Reliability and security

Two presses, one winner

A row lock serialises the buttons. The second person is told who got there first, and when.

One alert, no scheduler

The server sleeps when idle, so every board refresh runs the late-pickup check. An atomic update makes it fire once.

Named accounts

Salted scrypt, a signed session cookie, and an epoch that signs every device out on a password change.

Fail-closed admin

No admin list configured means nobody is an admin, the author included.

Allow-listed push targets

The server calls only known browser push services, so a subscription cannot point it anywhere else.

Safe Excel exports

A typed request number can never run as a formula in management's spreadsheet.

WHAT I LEARNED

Lessons from the floor

Installing is three problems

Chrome decides when to offer it, iPhones never do, and messenger browsers cannot. One in-app button now covers all three.

Fix the cause, not the phone

A phone kept losing notifications and clearing its data helped, twice. The cause was our sign-out cancelling the subscription. Now sign-out only tells the server.

A shared style touches every page

A header fix leaked into the login card for an hour. Now every page using a changed style is checked on a phone before release.

SUMMARY

The project in numbers

11	2	5	210
staff users	channels, one rulebook	part-count bands	automated tests

Python · FastAPI · PostgreSQL · vanilla JS PWA · Web Push · Google Cloud Run

givijananashvili40@gmail.com · linkedin.com/in/givi-jananashvili-a9877026b · github.com/Azazell